

Malware, or "Malicious software," is a type of software created and secretly injected into a system with the purpose of infiltrating, disrupting, or stealing information, or compromising the confidentiality, integrity, and availability of the victim's computer.

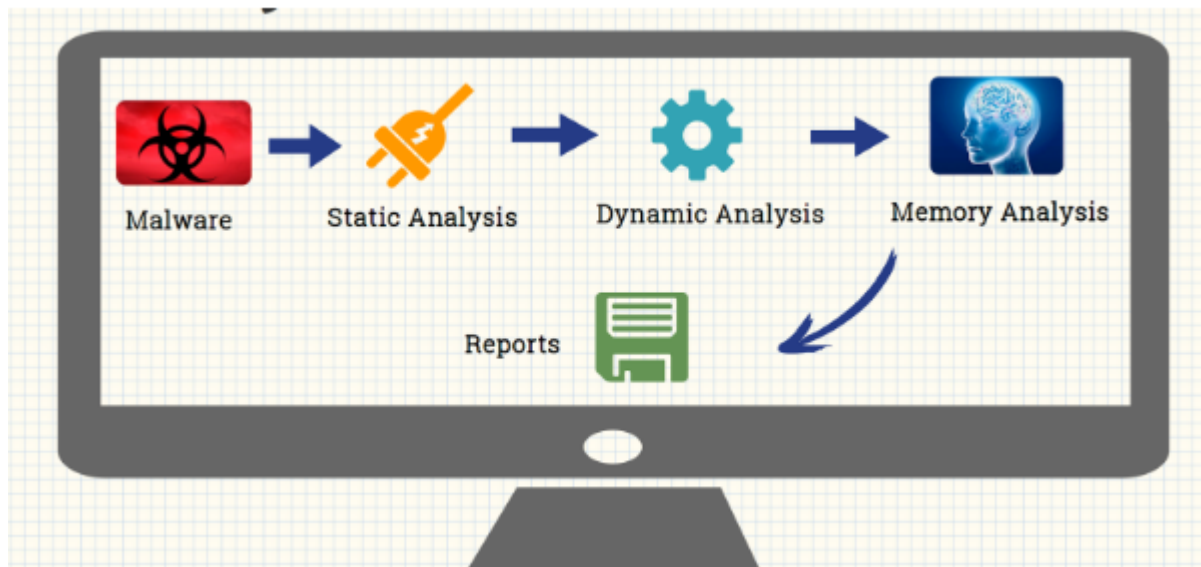
Malware is classified into many types depending on its function, method of infection, and destructive power: viruses, worms, trojans, rootkits, etc. To find malware, we need analytical methods and **malware analysis techniques**.

Malware analysis This is a crucial step in completely preventing and eliminating malware from computers and networks; restoring the network to its original state; and tracing the source of the attack.

First, it's necessary to accurately determine what happened to the entire system and find all the infected devices, files, and applications.

When analyzing malware, we need to accurately identify what the malware can do, how to detect it in the network system, and how to measure the damage it can cause. Most importantly, we need to find the characteristic patterns (identifiers – signatures) to identify the malware, eliminate it, and restore the system.

There are two basic methods for malware analysis: static analysis and dynamic analysis. Static analysis is a method of analyzing and checking for malware without running it. Dynamic analysis is a method where the malware is run during the analysis process. Both techniques are further classified into basic and advanced analysis.

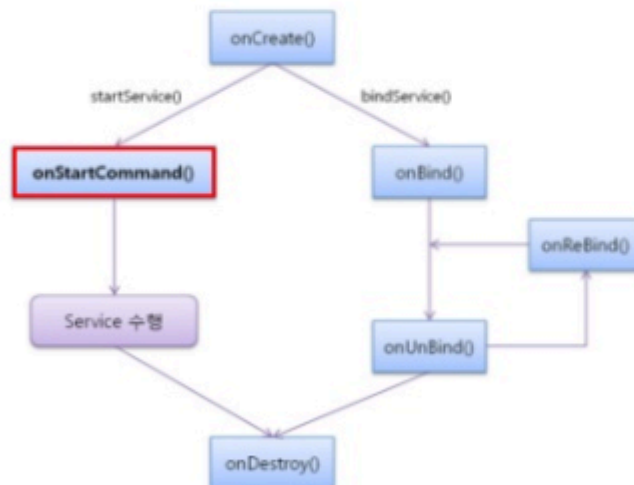


Basic static analysis

Basic static analysis involves examining executable files. Basic static analysis can identify a file as malicious, provide information about the functionality of that malicious file, and sometimes provide information that will allow you to create simple network signatures. Basic static analysis is simple and quick; however, it will not be effective against more sophisticated malware.

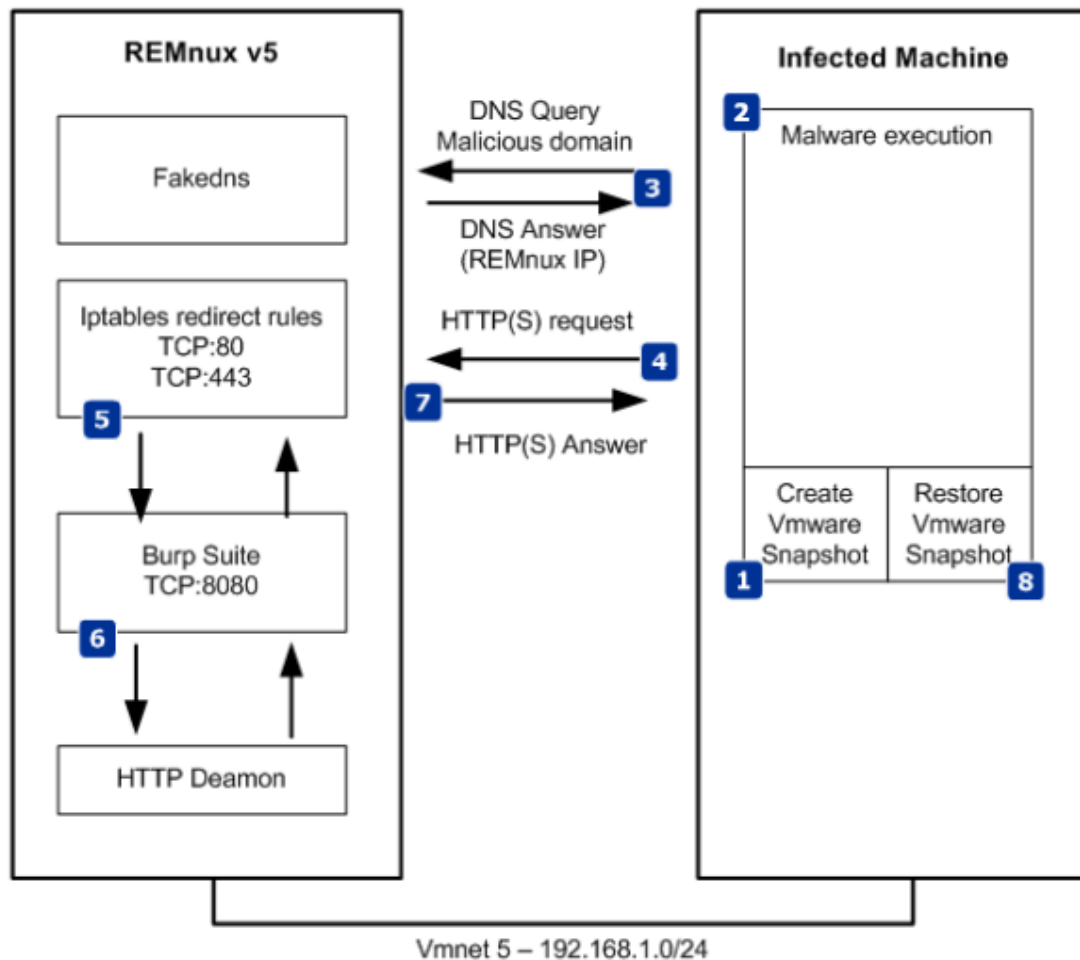
Static Analysis

- Service Life Cycle.



Fundamental analysis

Basic dynamic analysis methods involve running malware and observing its behavior on the system to eliminate infection, provide effective signatures, or both. However, before you can safely run malware, you must set up an environment that will allow you to study running malware without impacting your network. Similar to basic static analysis, basic dynamic analysis can be used by most people without requiring advanced programming knowledge; however, this will not be effective for all malware.



Advanced static analysis

Advanced static analysis (STA) methods involve reverse engineering the malware's internal contents by reading the executable file into a separator and examining the program's instructions to detect suspicious programs. Since the instructions are executed by the CPU, advanced STA will tell you which programs are suspicious. However, advanced STA requires specialized knowledge of programming, code structure, and Windows operating system concepts.

There are a few rules to keep in mind to make the analysis process more efficient.

1. We shouldn't focus too much on the details; most malware has a large amount of complex source code and contains a lot of junk code. Therefore, we should focus on the main features to avoid wasting time detailing unnecessary information.

2. Each analytical tool has its own functions, and it's necessary to apply them flexibly and effectively based on the strengths of each tool to make the analysis process faster and more accurate.
3. There are many different analytical strategies, so it's necessary to adapt strategies flexibly for better analytical results.
4. Job **malware analysis** Like a game of chase, the malware writer is always trying to hide their tracks, while the analyst is always trying to uncover the most detailed information. New techniques are created every day, so it's necessary to constantly update analytical techniques and technologies, and also to be creative to make the analysis process faster and more efficient.